

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the total cost and change.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase.

INPUT	PROCESS	OUTPUT
Numbers of notebook Number of pen Payment amount	Notebook cost = (3) x ₱45 Pen Cost = (2) x ₱12 Notebook cost + Pen cost = total cost Payment - total cost = change	Total cost Change

DECOMPOSITION

- Enter the number of notebooks.
- Enter the number of pens.
- Enter the payment amount.
- Calculate the notebook cost.
- Calculate the pen cost.
- Add the notebook cost and pen cost to get the total cost.
- Subtract the total cost from the payment to get the change.
- Display the total cost and change.

PSEUDOCODE

START

INPUT number of notebooks

INPUT number of pens

INPUT payment

notebookCost = number of notebook x 45

penCost = number of pens x 12

DISPLAY totalCost

DISPLAY change

END

DRY-RUN CALCULATION: QUICK ACTIVITY ONE

CASE TEST 1:

Input:

- Notebook: 3
- Pens: 2
- Payment: ₱200

STEP	VARIABLE	CALCULATION	RESULT
1	notebookCost	3 x ₱45	₱135
2	penCost	2 x ₱12	₱24
3	totalCost	₱135 + ₱24	₱159
4	change	₱200 - ₱159	₱41

CASE TEST 2:

Input:

- Notebook: 2
- Pens: 3
- Payment: ₱150

STEP	VARIABLES	CALCULATION	RESULT
1	notebookCost	2 x ₱45	₱90
2	penCost	3 x ₱12	₱36
3	totalCost	₱90 + ₱36	₱126
4	Change	₱150 - ₱126	₱24